

RISC-V Toolchain - Phases from C/C++ application code to memory-based binary files

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Phase 1 - Write and develop your HLL application using C/C++:

1. Option of using an online IDE:
[C/C++ online IDE](#)
2. Option of using local IDEs such as Visual Studio, CLion, etc:
[vs_WDExpress_2017.exe](#)

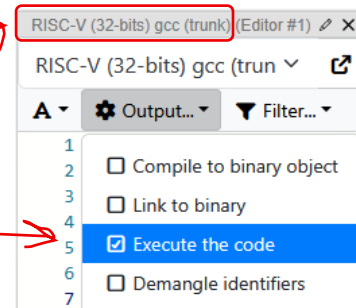
Phase 2 - Disassemble your C/C++ application using the RISC-V gcc-compiler:

1. Option of using an online gcc-compiler:
[C/C++ to RV32I disassembly online gcc-compiler](#)

Choose this gcc configuration

Choose the Execute mode

2. Option of using a manual compilation:
Write the equivalent assembly code yourself.



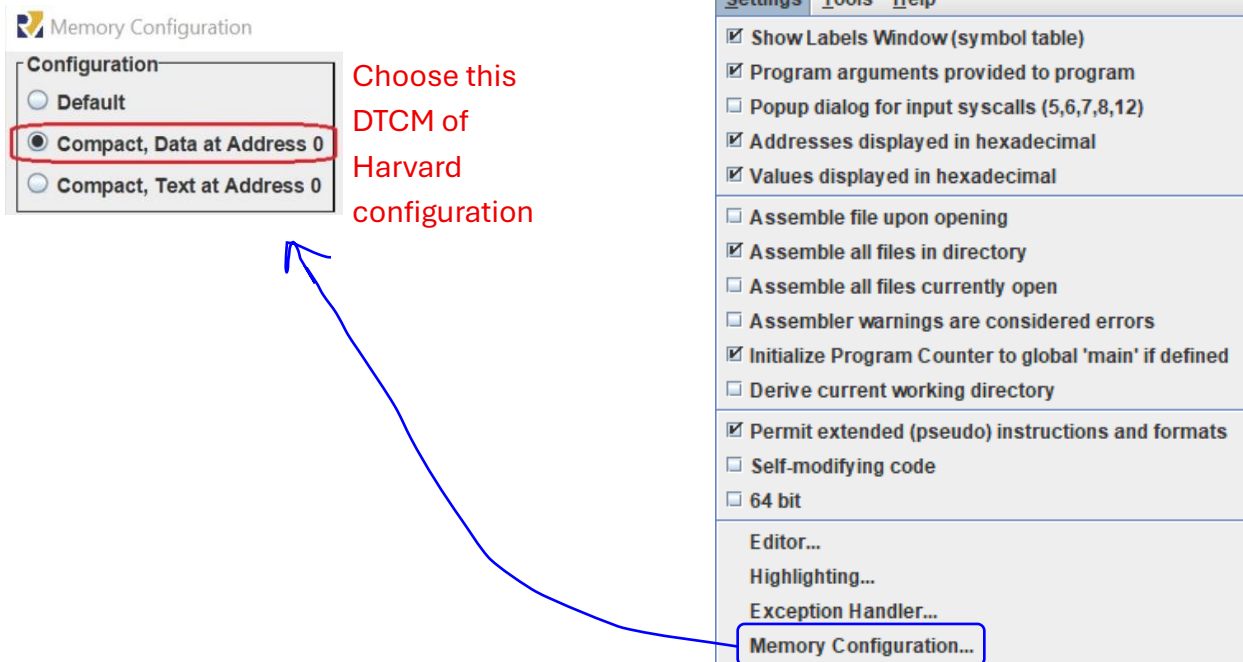
Note: In case the gcc disassembly code contains a system call to a standard C library function, we can't use the gcc-compiler for our unprivileged core (we need to compile the C/C++ code manually).

Phase 3 – Use assembly RARS IDE:

1. Open the RARS assembly IDE:

[RARS RISC-V Assembler, Simulator, and Runtime IDE \(local\)](#)

2. Make sure of the next RARS Configurations:



Note: For multiple assembly source files, prefix numbering is required according to their compilation order.

3. In case of using gcc disassembly (see Phase 2.1), make the following four changes:
 - i. Add two directives at the beginning of the data segment:
`.globl main`
`.data`
 - ii. Add the directives at the beginning of the code segment (before the label main):
`.text`
 - iii. Replace the directive `.zero` with `.space`
 - iv. Replace the assembly instruction name `lla` with `la`
4. Create from the assembly code files ITCM.h and DTCM.h of Hexadecimal-Text format.

Phase 4 – Create ITCM.hex and DTCM.hex execution binary files:

Use the following application (for Windows) to create binary execution files ITCM.hex and DTCM.hex suitable for our platform: [TextHex32bit-to-IntelHex32bit.exe](#)